3. Find the GCD of each pair of the following polynomials

(i) $12(x^4 - x^3)$, $8(x^4 - 3x^3 + 2x^2)$ whose LCM is $24x^3(x-1)(x-2)$

(ii) $(x^3 + y^3)$, $(x^4 + x^2y^2 + y^4)$ whose LCM is $(x^3 + y^3)(x^2 + xy + y^2)$

4. Given the LCM and GCD of the two polynomials $p(x)$ and $q(x)$ find the unknown polynomial in the following table

S.No.	LCM	GCD	$p(x)$	$q(x)$
(i)	$a^3 - 10a^2 + 11a + 70$	$a - 7$	$a^2 - 12a + 35$	
(ii)	$(x^4 - y^4)(x^4 + x^2y^2 + y^4)$	$(x^2 - y^2)$		$(x^4 - y^4)(x^2 + y^2 - xy)$

3.4 Rational Expressions

Definition : An expression is called a rational expression if it can be written in the form $\frac{p(x)}{q(x)}$ where $p(x)$ and $q(x)$ are polynomials and $q(x) \neq 0$. A rational expression is the ratio of two polynomials.

The following are examples of rational expressions.

$$\frac{9}{x}, \frac{2y+1}{y^2 - 4y + 9}, \frac{z^3 + 5}{z - 4}, \frac{a}{a + 10}.$$

The rational expressions are applied for describing distance-time, modeling multi-task problems, to combine workers or machines to complete a job schedule and much more.

3.4.1 Reduction of Rational Expression

A rational expression $\frac{p(x)}{q(x)}$ is said to be in its lowest form if $GCD\big(p(x), q(x)\big) = 1$.

To reduce a rational expression to its lowest form, follow the given steps

(i) Factorize the numerator and the denominator

(ii) If there are common factors in the numerator and denominator, cancel them.

(iii) The resulting expression will be a rational expression in its lowest form.

Example 3.13 Reduce the rational expressions to its lowest form

(i) $\frac{x-3}{x^2 - 9}$ (ii) $\frac{x^2 - 16}{x^2 + 8x + 16}$

Solution (i) $\frac{x-3}{x^2 - 9} = \frac{x-3}{(x+3)(x-3)} = \frac{1}{x+3}$

(ii) $\frac{x^2 - 16}{x^2 + 8x + 16} = \frac{(x+4)(x-4)}{(x+4)^2} = \frac{x-4}{x+4}$



3.4.2 Excluded Value

A value that makes a rational expression $\frac{p(x)}{q(x)}$ (in its lowest form) undefined is called an Excluded value.

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